

43)

$$\tau_p \frac{dy(t)}{dt} + y(t) = K_p f(t)$$

$$y(0) = 0, f(0) = 0$$

Taking laplace

$$\tau_p [sY(s) - y(0)] + Y(s) = K_p F(s)$$

$$\text{as } y(0) = 0$$

$$\tau_p s Y(s) + Y(s) = K_p F(s)$$

$$Y(s) = \frac{K_p F(s)}{1 + \tau_p s}$$

Transfer eq: $\frac{Y(s)}{F(s)} = \frac{K_p}{1 + \tau_p s}$